Multi-Vector Attack Correlation — Incident Report

Multi-Vector Attack Correlation — SSH Brute Force & Web Reconnaissance

Analyst: Daniel Agbosua · SOC Analyst **Investigation date:** 22-09-2026 · **Incident date:** 04-09-2026 **Environment:** LearnCyber (training/lab)

Executive summary

The investigation identified malicious activity against LearnCyber surfaces from the **195.178.110.0/24** network. This included repeated failed SSH authentication attempts against the `root` account from **195.178.110.228**, and a separate Cloudflare-blocked web request from **195.178.110.15** targeting `/wp-config.php-backup`.

The two events occurred approximately **3 seconds apart** (SSH authentication failure at 14:30:42; Cloudflare block at 14:30:45), and the source IPs share common infrastructure — the same ISP, **TECHOFF SRV LIMITED**, across adjacent ASNs **AS48089** (.228) and **AS48090** (.15, .30). The evidence supports **broader reconnaissance / multi-vector malicious activity, with medium confidence**. No successful SSH authentication or successful retrieval of the targeted web resource was identified.

Alert triage

Field	Value
Primary SSH source IP	195.178.110.228
Web attack source IP	195.178.110.15
Target system	soc
Target account	root
Authentication method	SSH
Authentication result	Failed
Rule description	PAM: user login failed
Severity	5 (individual PAM failure)

Host investigation

This was not a single failed login but multiple attempts across several minutes. It was **not** password spraying — a single `root` account was hit repeatedly, which confirms **brute force / password guessing**. The mechanics are clearly automated/bot-driven (rapid connect–fail–disconnect, sequential source ports, machine-like timing). **No compromise** was identified.

Incident timeline (Sep 04, BST)

Time	Source	IP	Activity
14:25:39	Attacker	195.178.110.228	Connection closed — port 33318 (initial probe)
14:28:57	Attacker	195.178.110.228	Authentication failure (user=root)
14:29:00	Attacker	195.178.110.228	Failed password for root — port 36664 (attempt 1)
14:29:01	Attacker	195.178.110.228	Connection closed [preauth] — port 36664
14:30:42	Attacker	195.178.110.228	Authentication failure (user=root)
14:30:44	Attacker	195.178.110.228	Failed password for root — port 39892 (attempt 2)
14:30:44	Attacker	195.178.110.228	Connection closed [preauth] — port 39892
14:30:45	Attacker	195.178.110.15	Cloudflare WAF blocked GET /wp-config.php-backup
14:32:32	Attacker	195.178.110.228	Authentication failure (user=root)
14:32:35	Attacker	195.178.110.228	Failed password for root — port 43078 (attempt 3)
14:32:36	Attacker	195.178.110.228	Connection closed [preauth] — port 43078
14:33:40	Analyst	localhost	Investigation: grep

Time	Source	IP	Activity
			195.178.110.228 / var/log/auth.log
14:34:16	Analyst	localhost	Investigation: grep 195.178.110.228 / var/log/auth.log
Post-incident	Analyst	localhost	Containment: ufw deny from 195.178.110.0/24 to any

Threat intelligence

Examined via VirusTotal, GreyNoise, AbuseIPDB, and Shodan. All three IPs: ISP **TECHOFF SRV LIMITED**, Netherlands, **100% AbuseIPDB confidence of abuse**.

IP	ASN	Vendors flagging	AbuseIPDB reports	Scanner categories	OS (Shodan)
195.178.110.228	AS48089	15/89	52,135 (1,775 sources)	Brute force, SSH, hacking, port scan, web app attack	Linux
195.178.110.30	AS48090	10/89	422,921 (596 sources)	Brute force, SSH, port scan, web app attack	Ubuntu
195.178.110.15	AS48090	14/89	178,665 (1,836 sources)	Brute force, hacking, port scan, bad web bot, web app attack	Linux

MITRE ATT&CK mapping

- **T1110.001 Password Guessing — Confirmed.** Repeated SSH password failures against root in auth.log.
- **T1595 Active Scanning — Potentially applicable.** Threat-intel associates the sources with scanning; not confirmed against LearnCyber telemetry.
- **T1190 Exploit Public-Facing Application — Potential, not confirmed.** The `/wp-config.php-backup` request was blocked; a probe is not proof of successful exploitation.

Containment & remediation

Contained via `ufw deny from 195.178.110.0/24`. Full SSH, web, monitoring, and long-term detection-engineering recommendations are documented separately (see `remediation.md`).

Final analyst verdict

Broader reconnaissance / multi-vector malicious activity against LearnCyber — medium confidence. The temporal, behavioural, and infrastructure correlations are meaningful and rule out an isolated event, but the evidence does not conclusively prove a single threat actor. **No successful SSH compromise and no successful retrieval of the WordPress backup file were identified in the reviewed evidence.**